

Usage benefits	Example command
Visualises real-time traffic	<i>nload</i>
Monitors specific interfaces	<i>nload eth0</i>
Tracks total traffic data	<i>nload -m</i>

By combining interface management and real-time monitoring, administrators can maintain a well-functioning network and promptly address any disruptions or inefficiencies.

Traffic analysis tools

Traffic analysis tools are essential for understanding and optimising network traffic. These tools allow administrators to capture, inspect, and analyse packets while also examining socket and port activity. Such analysis ensures network efficiency, identifies potential issues, and supports security monitoring.

Key questions administrators face daily with respect to network traffic are:

- Are there anomalies or unexpected patterns in network traffic?
- Are the correct ports open and secure for essential services?

Packet capture and inspection

tcpdump: tcpdump is a command-line packet analyser that captures network traffic for detailed inspection. It is widely used for troubleshooting and monitoring network performance.

Usage benefits	Example command
Captures packets on a specific interface	<i>sudo tcpdump -i eth0</i>
Filters traffic by IP	<i>sudo tcpdump host 192.168.1.1</i>
Saves captured packets to a file	<i>sudo tcpdump -w capture.pcap</i>

tshark (Wireshark CLI): tshark is the command-line version of Wireshark, allowing packet capture and inspection in a lightweight format.

Usage benefits	Example command
Captures packets for analysis	<i>sudo tshark -i eth0</i>
Displays packets live	<i>sudo tshark</i>
Saves captures to a file	<i>sudo tshark -w output.pcap</i>

Socket and port analysis

ss: ss provides detailed insights into open sockets, connections, and listening ports, replacing the older netstat command.

Usage benefits	Example command
Lists all active connections	<i>ss -tuln</i>
Displays detailed connection stats	<i>ss -s</i>
Filters connections by port	<i>ss -l sport = :22</i>

nmap: nmap (Network Mapper) is a powerful tool for network discovery and security auditing. It scans for open ports, identifies services, and detects vulnerabilities.

Usage benefits	Example command
Scans for open ports	<i>nmap -p 1-65535 localhost</i>
Detects OS and services	<i>nmap -A 192.168.1.1</i>
Performs a stealth scan	<i>nmap -sS 192.168.1.1</i>

Traffic analysis tools like tcpdump, tshark, ss, and nmap provide essential insights into network traffic and security. They enable administrators to identify bottlenecks, optimise performance, and safeguard against threats, ensuring efficient and secure network operations.

File transfer and remote access tools

File transfer and remote access are integral to network administration. These tools ensure secure and efficient data movement across systems and allow administrators to manage servers remotely. Reliable file transfer solutions and secure remote access mechanisms are essential for maintaining operational continuity.

Key questions administrators face daily with respect to file transfer and remote access are:

- Are files being transferred securely between systems?
- Is remote server access secured against unauthorised users?

Secure file transfer

scp: scp (Secure Copy) is a command-line tool that transfers files securely between systems over SSH. It ensures data protection during transit.

Usage benefits	Example command
Transfers files securely	<i>scp file.txt user@192.168.1.1:/path</i>
Copies directories recursively	<i>scp -r folder user@remote:/path</i>
Transfers files with a specific port	<i>scp -P 2222 file user@host:/path</i>

rsync: rsync synchronises files and directories between systems, using delta transfers to minimise bandwidth usage. It is a highly efficient tool for backups and file synchronisation.